



Randomization

Rehan Naeem

Randomized
Transaction

Monitor
Modification

Constraints

Weighted
Distribution

Concurrent
Constraint
Evaluation

Implication
and
Equivalence
Operators

Signed and
Unsigned
Variable Issue

Randomization

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Outline

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- 1 Randomized Transaction
- 2 Monitor Modification
- 3 Constraints
- 4 Weighted Distribution
- 5 Concurrent Constraint Evaluation
- 6 Implication and Equivalence Operators
- 7 Signed and Unsigned Variable Issue



Randomized Transaction

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Signed and Unsigned Variable Issue

```
class Transaction;
  static int count;           // static variable
  int id;                    // nonstatic variable

  rand bit [3:0] a;
  rand bit [3:0] b;
  bit [3:0] c;

  function new(bit [3:0] a, bit[3:0] b, bit[4:0] c);
    this.a = a;
    this.b = b;
    this.c = c;
    id = count;
    count = count+1;
  endfunction

  function display();
    $display("-----");
    $display("count=%d, ID=%d, a = %d, b=%d, c=%d", count, id, a, b, c);
    $display("-----");
  endfunction
endclass
```

(a) Transaction Code



Randomized Transaction

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Signed and Unsigned Variable Issue

```
class Transaction;
static int count;           // static variable
int id;                     // nonstatic variable
rand bit [3:0] a;
rand bit [3:0] b;
bit [3:0] c;

function new(bit [3:0] a, bit [3:0] b, bit [4:0] c);
    this.a = a;
    this.b = b;
    this.c = c;
    id = count;
    count = count+1;
endfunction

function display();
    $display("-----");
    $display("count=%d, ID=%d, a = %d, b=%d, c=%d", count, id, a, b, c);
    $display("-----");
endfunction
endclass
```

(a) Transaction Code

```
class transaction;
static int count;
int id;
rand bit [3:0] a;
rand bit [3:0] b;
bit [4:0] c;

function new();
    id = count;
    count++;
endfunction

function display(string caller);
    $display("%%s-Transaction # %0d, Total Transactions = %0d", caller, id, count);
    $display("|- A = %2d, B=%2d, C=%2d\n", a, b, c);
endfunction
endclass
```

(b) Modified Transaction Code



Monitor Modification

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```
class transaction;
    static int count;
    int id;
    rand bit [3:0] a;
    rand bit [3:0] b;
    bit [4:0] c;

    function new();
        id = count;
        count++;
    endfunction

    * function set_values(bit [3:0] a, b, c);
        this.a = a;
        this.b = b;
        this.c = c;
    endfunction

    function display(string caller);
        $display("%s-Transaction # %0d, Total Transactions = %0d", caller, id, count);
        $display("[*]- A = %2d, B=%2d, C=%2d\n", a, b, c);
    endfunction
endclass
```

(a) Transaction Code for Monitor

```
class monitor;
    mailbox mon2scb;
    int repeat_count;
    virtual ra_intf.mon rinf;
    function new(mailbox mon2scb, int repeat_count, virtual ra_intf.mon
rinf);
        this.mon2scb = mon2scb;
        this.repeat_count = repeat_count;
        this.rinf = rinf;
    endfunction

    task run();
        transaction trans;
        int ncount = 0;
        bit [3:0] a, b;
        while(ncount < repeat_count) begin
            @(posedge rinf.clk);
            if(!rinf.reset && rinf.valid) begin
                a = rinf.a;
                b = rinf.b;
                @(posedge rinf.clk);
                trans = new();
                trans.set_values(a, b, rinf.c);
                mon2scb.put(trans);
                trans.display("[Monitor]");
                ncount++;
            end
        end
    endtask
endclass
```

(b) Modified Monitor Code



Randomized Transaction

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Note

- rand vs randc



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Note

- rand vs randc
- monitor code: A method instead of constructor for packet values



Constraints - Example 1

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Variable Issue

```
class transaction;
  static int count;
  int id;
  rand bit [3:0] a;
  rand bit [3:0] b;
  bit [4:0] c;

  constraint bcd {
    a >= 0;
    a <= 9;
    b >= 0;
    b <= 9;
  }

  function new();
    id = count;
    count++;
  endfunction

  function set_values(bit [3:0] a, b, c);
    this.a = a;
    this.b = b;
    this.c = c;
  endfunction

  function display(string caller);
    $display("s-Transaction # %0d, Total Transactions = %0d", caller, id, count);
    $display("|- A = %2d, B=%2d, C=%2d\n", a, b, c);
  endfunction
endclass
```

(a) Transaction Code with Constraints



Constraints - Example 2

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Variable Issue

```
class Child;  
    bit [7:0] age; // Error - should be rand or randc  
    constraint c_teenager {age > 12;  
                           age < 20;}  
endclass
```



Constraints - Example 3

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Variable Issue

```
class Stim;
  const bit [31:0] CONGEST_ADDR = 42;
  typedef enum {READ, WRITE, CONTROL} stim_e;
  randc stim_e kind;    // Enumerated var
  rand bit [31:0] len, src, dst;
  rand bit congestion_test;

  constraint c_stim {
    len < 1000;
    len > 0;
    if (congestion_test) {
      dst inside {[CONGEST_ADDR-10:CONGEST_ADDR+10]};
      src == CONGEST_ADDR;
    }
    else
      src inside {0, [2:10], [100:107]};
  }
endclass
```



Constraints - Example 3

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Variable Issue

Alert

Be careful with ==



Constraints - Example 4

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Variable Issue

```
class Order_bad;
    rand bit [7:0] lo, med, hi;
    constraint bad {lo < med < hi;} // Gotcha!
endclass
```

(a) Bad Code



Constraints - Example 4

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Variable Issue

```
class Order_bad;
    rand bit [7:0] lo, med, hi;
    constraint bad {lo < med < hi;} // Gotcha!
endclass
```

(a) Bad Code

```
class Order_good;
    rand bit [7:0] lo, med, hi;
    constraint good {lo < med;    // Only use binary constraints
                    med < hi;}
endclass
```

(b) Good Code



Array in Constraints

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Variable Issue

```
class Fib;  
    rand bit [7:0] f;  
    bit [7:0] vals[] = '{1,2,3,5,8};  
    constraint c_fibonacci {  
        f inside vals;  
    }  
endclass
```



Weighted Distribution - Example 1

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Variable Issue

```
class Transaction;
  rand bit [1:0] src, dst;
  constraint c_dist {
    src dist {0:=40, [1:3]:=60};
    // src = 0, weight = 40/220
    // src = 1, weight = 60/220
    // src = 2, weight = 60/220
    // src = 3, weight = 60/220

    dst dist {0:/40, [1:3]:/60};
    // dst = 0, weight = 40/100
    // dst = 1, weight = 20/100
    // dst = 2, weight = 20/100
    // dst = 3, weight = 20/100
  }
endclass
```

(a) Class with Weights



Weighted Distribution - Example 2

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```
// Bus operation, byte, word, or longword
class BusOp;
    // Operand length
    typedef enum {BYTE, WORD, LWRD } length_e;
    rand length_e len;

    // Weights for dist constraint
    bit [31:0] w_byte=1, w_word=3, w_lwrld=5;

    constraint c_len {
        len dist {BYTE := w_byte,          // Choose a random
                  WORD := w_word,          // length using
                  LWRD := w_lwrld};        // variable weights
    }
endclass
```

(a) Class with Weights



Concurrent Constraint Evaluation

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```
class Bidir;
  rand bit [15:0] r, s, t;
  constraint c_bidir {           // All are solved in parallel
    r < t;                       // A value for r affects s, t
    s == r;
    t < 10;
    s > 5;
  }
endclass
```

Solutions for bidirectional constraint

<i>Solution</i>	<i>r</i>	<i>s</i>	<i>t</i>
A	6	6	7
B	6	6	8
C	6	6	9
D	7	7	8
E	7	7	9
F	8	8	9



Implication Operator

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Variable Issue

```
class BusOp;  
  rand bit [31:0] addr;  
  rand bit io_space_mode;  
  constraint c_io {  
    io_space_mode ->  
      addr[31] == 1'b1;  
  }  
}
```

Implication operator truth table

$A \rightarrow B$	$B = \text{false}$	$B = \text{true}$
A=false	true	true
A=true	false	true



Equivalence Operator

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```
rand bit d, e;  
constraint c { (d==1) <-> (e==1); }
```

Equivalence operator truth table

$A \leftrightarrow B$	$B = \text{false}$	$B = \text{true}$
$A = \text{false}$	true	false
$A = \text{true}$	false	true



Implication Operator - Example

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Variable Issue

```
class Imp1;
    rand bit x;           // 0 or 1
    rand bit [1:0] y;     // 0, 1, 2, or 3
    constraint c_xy {
        (x==0) -> (y==0);
    }
endclass
```



Implication Operator - Example

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Variable Issue

```
class Imp1;  
    rand bit x;           // 0 or 1  
    rand bit [1:0] y;     // 0, 1, 2, or 3  
    constraint c_xy {  
        (x==0) -> (y==0);  
    }  
endclass
```

Table 6.5 Solutions for Imp1 class

<i>Solution</i>	<i>x</i>	<i>y</i>	<i>Probability</i>
A	0	0	1/2
B	0	1	0
C	0	2	0
D	0	3	0
E	1	0	1/8
F	1	1	1/8
G	1	2	1/8
H	1	3	1/8



Implication with Additional Constraint

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```
class Imp2;
  rand bit x;           // 0 or 1
  rand bit [1:0] y;     // 0, 1, 2, or 3
  constraint c_xy {
    y > 0;               // Force y = 1, 2, or 3
    (x==0) -> (y==0);
  }
endclass
```



Implication with Additional Constraint

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Variable Issue

```
class Imp2;
  rand bit x;           // 0 or 1
  rand bit [1:0] y;     // 0, 1, 2, or 3
  constraint c_xy {
    y > 0;              // Force y = 1, 2, or 3
    (x==0) -> (y==0);
  }
endclass
```

Table 6.6 Solutions for Imp2 class

<i>Solution</i>	<i>x</i>	<i>y</i>	<i>Probability</i>
A	0	0	0
B	0	1	0
C	0	2	0
D	0	3	0
E	1	0	0
F	1	1	1/3
G	1	2	1/3
H	1	3	1/3



Guiding Distribution

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```
class SolveXBeforeY;
  rand bit x;           // 0 or 1
  rand bit [1:0] y;     // 0, 1, 2, or 3
  constraint c_xy {
    (x==0) -> y==0;
    solve x before y;
  }
endclass
```

```
class SolveXBeforeY;
  rand bit x;           // 0 or 1
  rand bit [1:0] y;     // 0, 1, 2, or 3
  constraint c_xy {
    (x==0) -> y==0;
    solve y before x;
  }
endclass
```

<i>Solution</i>	<i>x</i>	<i>y</i>	<i>Probability</i>
A	0	0	1/2
B	0	1	0
C	0	2	0
D	0	3	0
E	1	0	1/8
F	1	1	1/8
G	1	2	1/8
H	1	3	1/8

<i>Solution</i>	<i>x</i>	<i>y</i>	<i>Probability</i>
A	0	0	1/8
B	0	1	0
C	0	2	0
D	0	3	0
E	1	0	1/8
F	1	1	1/4
G	1	2	1/4
H	1	3	1/4



Randomization with Signed Variables

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Variable Issue

```
class SignedVars;  
    rand byte pkt1_len, pkt2_len;  
    constraint total_len {  
        pkt1_len + pkt2_len == 64;  
    }  
endclass
```



Randomization with Unsigned Variables

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Variable Issue

```
class Vars32;  
    rand bit [31:0] pkt1_len, pkt2_len; // unsigned type  
    constraint total_len {  
        pkt1_len + pkt2_len == 64;  
    }  
endclass
```



Randomization with Unsigned Variables

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```
class Vars8;
  rand bit [7:0] pkt1_len, pkt2_len;    // 8-bits wide
  constraint total_len {
    pkt1_len + pkt2_len == 9'd64;      // 9-bit sum
  }
endclass
```



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